Table 1. Illustrative Examples of Survey-to-OMOP Concept Mapping

Use Case (Row)	Source Survey Item / Response	OMOP Domain	Domain Concept Field: ID	Qualifier Concept ID	Type / Value Concept ID
One-to-one (1)	Marital status; response "Married" (source: SNOMED CT 125680007)	Observation	*observation_concept_id*: 4053609 (Marital status)	–	*value_as_concept_id*: 4338692 (Married)
One-to-one (2)	Troponin T (source: LOINC 67151-1, Troponin T.cardiac [Mass/volume] in Serum or Plasma by High sensitivity method)	Measurement	*measurement_concept_id*: 40769783	–	*value_as_number*: measured result
Question–response pair; negative response (3)	"Has a doctor or other healthcare professional ever told you that you have/had…"; condition: Myocardial infarction	Observation	*observation_concept_id*: 2005200014	4329847 (Myocardial infarction)	*observation_type_concept_id*: 32862 (Survey); *value_as_concept_id*: 4188540 (No) or 36311156 (Prefer not to say)
Question–response pair; dual-domain (4)	Same question; condition: Type 2 diabetes mellitus	Observation	*observation_concept_id*: 2005200014	201826 (Type 2 diabetes mellitus)	*observation_type_concept_id*: 32862 (Survey); *value_as_concept_id*: 4188539 (Yes)
Dual-domain (5)	Same question and response as row 4	Condition	*condition_concept_id*: 201826 (Type 2 diabetes mellitus)	–	*condition_type_concept_id*: 45905770 (Patient reported)

Representative examples of the four mapping use cases described in the text. Rows 1–2 show one-to-one mapping to existing standard concepts. Rows 3–4 show question–response pair mapping, in which the question is recorded as the observation concept, the referenced condition in qualifier_concept_id, and the response in value_as_concept_id; row 3 also illustrates negative-response capture. Rows 4–5 show dual-domain mapping: a positive response to a professional-diagnosis question generates both an Observation record and a Condition record with a patient-reported type concept. Concept IDs ≥2,000,000,000 are AI-READI custom extension concepts.